

Wasserstein K -Center Clustering with Robust Validation Metrics and Consistency Guarantees

You

November 27, 2025

Abstract

Contemporary data in many scientific domains, such as images, media streams, omics, are naturally represented as probability distributions in Wasserstein space instead of points in Euclidean space. Clustering methods for distribution-valued data are in high demand, as they provide exploratory frameworks to group objects with similarity. The classical K-means clustering, which groups the data points based on their relative distance, seems to have a natural extension to Wasserstein space via replacing the Euclidean distance by Wasserstein distance. The curvature and lack of linear structure inherent to Wasserstein geometry complicate extensions of Euclidean K -center methods, both in defining cluster centers and in selecting the number of clusters.

When the objects to be clustered are distributions, the Wasserstein space provides a geometrically meaningful setting that respects the underlying mass-transportation structure of the data. k-means clustering in Wasserstein space leverages this geometry by grouping distributions according to their Wasserstein distances, allowing clusters to reflect differences in both location and shape. Distribution-valued data appear in an increasing range of modern applications, which makes clustering in the Wasserstein space $\mathcal{P}_2(\mathbb{R}^d)$ an important problem. The curvature and lack of linear structure inherent to Wasserstein geometry complicate extensions of Euclidean K -center methods, both in defining cluster centers and in selecting the number of clusters.

A Wasserstein K -center framework is developed using barycentric computation, Wasserstein Fréchet medians, and a cluster-separation metric based on relative Wasserstein spatial depth (ReD). This metric is intrinsically defined on $\mathcal{P}_2(\mathbb{R}^d)$, robust to distributional heterogeneity, and provides a stable criterion for tuning K . A Pollard-type strong consistency theorem is established for Wasserstein K -means, and experiments on synthetic mixtures, MNIST distributional representations, and AML cytometry data demonstrate the effectiveness of the resulting approach.

1 Introduction

Contemporary data is complex and multi-faceted, often representing heterogeneous populations, functional observations, or spatially dependent structures. Rather than modeling each data object as a single vector $x \in \mathbb{R}^d$, it is often more appropriate to represent it as a probability distribution $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, where $\mathcal{P}_2(\mathbb{R}^d)$ denotes the set of Borel probability measures on $\mathbb{R}^d$ with finite second moments. This formulation captures intrinsic variability and uncertainty in a geometrically meaningful way, allowing data analysis to be carried out directly in the *Wasserstein space* equipped with the quadratic Wasserstein distance $\mathcal{W}_2$ [1, 2]. Let $\mathbf{P} \in \mathcal{P}(\mathcal{P}_2(\mathbb{R}^d))$ denote a *distribution of distributions*, that is, a probability measure whose elements are themselves distributions in the Wasserstein space. Our objective is to analyze the structure of $\mathbf{P}$ by identifying coherent clusters among its constituent distributions.

Existing clustering methods in Wasserstein space. Clustering is a fundamental unsupervised learning task that seeks to partition an unlabeled dataset into groups of similar objects according to a specified metric or similarity function [3]. In the Wasserstein space, this task has been addressed primarily through extensions of classical Euclidean algorithms. The *Wasserstein K -means* method [4] generalizes the standard K -means objective to distribution-valued data by minimizing

$$\min_{\{\nu_k, S_k\}_{k=1}^K} \sum_{k=1}^K \sum_{\mu_i \in S_k} \mathcal{W}_2^2(\mu_i, \nu_k), \quad (1)$$

where ν_k denotes the Wasserstein barycenter of cluster S_k . An alternative to centroid-based methods is to apply spectral clustering directly on the Wasserstein space, as recently formalized by Wang, Qiao, and Fan [5]. Given empirical distributions $\{\mu_i\}_{i=1}^n \subset \mathcal{P}_2(\Omega)$, pairwise dissimilarities are first computed using the squared Wasserstein distance $d_{ij} = \mathcal{W}_2^2(\mu_i, \mu_j)$. These distances are then transformed into a positive semi-definite affinity kernel through a Gaussian mapping

$$A_{ij} = \exp(-d_{ij}/\sigma^2), \quad (2)$$

where $\sigma > 0$ controls the neighborhood scale. This affinity matrix captures the local geometric relations among the distributions in Wasserstein space while preserving their geodesic structure.

From A , the normalized graph Laplacian is constructed as

$$L_{\text{sym}} = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(A\mathbf{1}),$$

and the embedding of the data is obtained from the first K eigenvectors of L_{sym} corresponding to its smallest eigenvalues. Finally, a standard Euclidean K -means is applied to the rows of the eigenvector matrix to recover cluster assignments.

This approach, by combining optimal transport geometry with spectral graph theory, enables clustering of complex and distributional data without requiring explicit barycenter computation, and has demonstrated superior performance in applications involving high-dimensional probability measures [5]. These approaches exploit the rich geometry of Wasserstein space and have demonstrated success in applications such as image analysis, functional data clustering, and flow cytometry [6, 7, 8].

Motivation and limitations. A persistent challenge in practice is that the true number of clusters K is typically unknown [9, 10]. In many real-world applications, researchers often lack reliable tools to determine whether a clustering result reflects meaningful structure or is merely an artifact of initialization or noise [11]. To address this, a variety of internal validation indices have been developed, each quantifying the quality of a clustering solution based on the relationships among the data points.

One of the most widely used measures is the *silhouette score* [12], which evaluates how similar each point is to its own cluster compared to other clusters. For a distribution μ_i belonging to cluster S_{k_i} , let

$$a_i = \frac{1}{|S_{k_i}| - 1} \sum_{\mu_j \in S_{k_i} \setminus \{\mu_i\}} \mathcal{W}_2(\mu_i, \mu_j)$$

denote the average within-cluster dissimilarity, and

$$b_i = \min_{\ell \neq k_i} \frac{1}{|S_\ell|} \sum_{\mu_j \in S_\ell} \mathcal{W}_2(\mu_i, \mu_j)$$

the minimum average distance from μ_i to all other clusters. The silhouette coefficient for μ_i is then given by

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}, \quad s_i \in [-1, 1], \quad (3)$$

where larger s_i indicates better separation from neighboring clusters. In practice, one computes the mean silhouette $\bar{s}(K) = \frac{1}{n} \sum_{i=1}^n s_i$ for different candidate K values, and selects $\hat{K} = \arg \max_K \bar{s}(K)$. Despite its simplicity, the silhouette measure is sensitive to overlapping or nonconvex clusters, as it relies on global averages of pairwise distances [13].

To alleviate the limitations of global averaging, a *local silhouette* variant [14] restricts the computation of a_i and b_i to the k -nearest neighbors of μ_i , yielding a more localized assessment of cohesion and separation. Formally, let $N_k(i)$ be the k nearest neighbors of μ_i within its cluster S_{k_i} and $N_k(i; \ell)$ the k nearest neighbors of μ_i within another cluster S_ℓ . One can define

$$a_i^{(k)} = \frac{1}{|N_k(i)|} \sum_{\mu_j \in N_k(i)} \mathcal{W}_2(\mu_i, \mu_j), \quad b_i^{(k)} = \min_{\ell \neq k_i} \frac{1}{|N_k(i; \ell)|} \sum_{\mu_j \in N_k(i; \ell)} \mathcal{W}_2(\mu_i, \mu_j),$$

and obtain a k -dependent local silhouette

$$s_i^{(k)} = \frac{b_i^{(k)} - a_i^{(k)}}{\max\{a_i^{(k)}, b_i^{(k)}\}}.$$

Model selection then proceeds analogously by maximizing the mean local silhouette $\bar{s}^{(k)}(K) = \frac{1}{n} \sum_i s_i^{(k)}(K)$. However, the choice of k remains heuristic, and the metric can still be destabilized by outliers and noisy boundary points [13].

A second classical metric is the *Davies–Bouldin Index (DBI)* [15], which evaluates clustering quality through the ratio between within-cluster scatter and between-cluster separation. For a cluster S_k with representative $\hat{\mu}_k$ (e.g. a Wasserstein barycenter or K-median), define its scatter as

$$R_k = \frac{1}{|S_k|} \sum_{\mu_i \in S_k} \mathcal{W}_2(\mu_i, \hat{\mu}_k),$$

and for any two clusters k and ℓ define the distance between representatives as

$$D_{k\ell} = \mathcal{W}_2(\hat{\mu}_k, \hat{\mu}_\ell).$$

DBI is then given by

$$\text{DBI} = \frac{1}{K} \sum_{k=1}^K \max_{\ell \neq k} \frac{R_k + R_\ell}{D_{k\ell}},$$

where smaller values indicate tighter and more separated clusters. Despite its simplicity, DBI is known to be highly sensitive to outliers and boundary noise, as the scatter term R_k increases sharply in their presence [13, 16]. It is also unstable under changes in cluster scale or variance [17], and provides only cluster-level values, limiting interpretability for heterogeneous or distribution-valued clusters.

The *Gap Statistic* [18] offers another popular approach for selecting K by comparing the observed within-cluster dispersion to that obtained under a reference distribution with “no clustering structure.” Let W_K denote a within-cluster dispersion measure for a K -partition, for instance

$$W_K = \sum_{k=1}^K \frac{1}{|S_k|} \sum_{\mu_i \in S_k} \mathcal{W}_2^2(\mu_i, \hat{\mu}_k),$$

and let $W_K^{*(b)}$ be the corresponding quantity for the b -th dataset drawn from a reference distribution. The Gap value is defined as

$$\text{Gap}(K) = \frac{1}{B} \sum_{b=1}^B \log W_K^{*(b)} - \log W_K,$$

and one typically selects the smallest K such that

$$\text{Gap}(K) \geq \text{Gap}(K+1) - s_{K+1},$$

where s_{K+1} is an estimate of the simulation variability. However, in the Wasserstein space, constructing a suitable reference distribution is highly nontrivial: without strong parametric assumptions, there is no canonical notion of uniformity or volume for probability measures, so a “null” family of distributions with no clustering structure is fundamentally ambiguous [19].

Beyond centroid- and distance-based methods, Density-Based Cluster Validation (DBCV) [20] evaluates clustering quality using the notion of density connectivity, capturing both the internal compactness and external separation of clusters. For each pair of points (μ_i, μ_j) , the mutual reachability distance is defined as

$$d_{\text{mr}}(\mu_i, \mu_j) = \max \{c_k(\mu_i), c_k(\mu_j), \mathcal{W}_2(\mu_i, \mu_j)\},$$

where $c_k(\mu)$ is the distance from μ to its k th nearest neighbor, serving as a proxy for local inverse density. Based on these distances, a similarity graph (e.g. a minimum spanning tree) is constructed, and each cluster’s density sparseness (the maximum internal edge) and density separation (the minimum inter-cluster edge) are computed. The validity of a cluster C is expressed as

$$V(C) = \frac{\text{separation}(C) - \text{sparseness}(C)}{\max\{\text{separation}(C), \text{sparseness}(C)\}},$$

and the overall index is the size-weighted average

$$\text{DBCV} = \sum_C \frac{|C|}{n} V(C).$$

Similar to silhouette-based procedures, the optimal number of clusters is chosen as $\hat{K} = \arg \max_K \text{DBCv}(K)$. While DBCv is well-suited for nonconvex or variable-density clusters, it depends strongly on user-specified parameters such as the neighborhood size k , and produces cluster-level rather than point-level scores [21].

In summary, although these validation measures can be adapted to the Wasserstein setting by replacing Euclidean distances with $\mathcal{W}_2$, they remain limited in robustness and interpretability for distribution-valued data. Silhouette and its local variants rely on averaged pairwise distances, which blur boundary information and are sensitive to noise; DBI is unstable under outliers and scale differences and only yields cluster-level summaries; the Gap Statistic requires a reference distribution that is difficult to define without parametric assumptions in the Wasserstein space; and DBCv depends on density thresholds and connectivity parameters and likewise lacks pointwise interpretability. These limitations motivate the development of a robust, interpretable, and pointwise validation statistic tailored specifically to the geometry of the Wasserstein space.

Our contribution. To address this issue, we introduce a new validation statistic for clustering in Wasserstein space based on the *Wasserstein spatial depth (WSD)* [2]. Data depth provides a center-outward measure of centrality that is affine-invariant and robust to noise [22, 23]. Extending this notion to Wasserstein space allows us to evaluate the relative centrality of each empirical distribution within and across clusters. The proposed WSD-based criterion serves as a robust validation tool that quantifies clustering reliability without requiring additional parameters or convexity assumptions. By combining the interpretability of depth functions with the geometric expressiveness of the Wasserstein metric, our approach provides a principled framework for assessing clustering quality and determining the optimal number of clusters in distribution-valued datasets.

2 Methodology

We propose a distributional clustering framework built upon optimal transport and Wasserstein geometry. The pipeline integrates four key components in a coherent hierarchy. First, a *semi-free support entropic barycenter* algorithm based on log-domain Sinkhorn iterations is developed to compute smooth and scalable Wasserstein barycenters. Second, this barycenter solver is embedded within an *IRLS (iteratively reweighted least squares)* scheme to compute Wasserstein Fréchet medians. Third, the median computation serves as the core subroutine of a *Wasserstein K-median clustering* algorithm, where global robustness is enhanced by simulated annealing and convergence is accelerated by a PAM-based warm start. Finally, clustering performance and model selection are validated using a *Wasserstein spatial depth*-based criterion: the *relative depth (W-ReD)* index quantifies within- versus between-cluster centrality, and the optimal cluster number K^* is chosen by maximizing the average W-ReD over all samples.

2.1 Semi-Free Support Barycenter for Empirical Distributions

Let $\{Y_i\}_{i=1}^n$ be a collection of empirical measures on $\mathbb{R}^d$, where each $Y_i = \{y_{ij}\}_{j=1}^{m_i}$ represents a discrete point cloud with weights b_{ij} . The Wasserstein barycenter problem seeks a measure ν^* that minimizes the average squared Wasserstein distance to all inputs:

$$\nu^* = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \frac{1}{n} \sum_{i=1}^n \mathcal{W}_2^2(\nu, \mu_i). \quad (4)$$

When the barycenter is allowed to have completely free support, the discrete barycenter problem admits a linear programming (LP) formulation over a transportation polytope that couples the barycenter with each input measure. In particular, if $\mu_i = \sum_{j=1}^{m_i} b_{ij} \delta_{y_{ij}}$ and the barycenter is written as $\nu = \sum_{p=1}^K a_p \delta_{x_p}$ with unknown supports $X = \{x_p\}_{p=1}^K$ and weights $a \in \Delta^K$, then one can introduce transport plans $T^{(i)} \in \mathbb{R}_+^{K \times m_i}$ satisfying $T^{(i)} \mathbf{1} = a$ and $(T^{(i)})^\top \mathbf{1} = b_i$ and minimize the total cost $\sum_i \langle T^{(i)}, M(X, Y_i) \rangle$. A key structural fact is the sparsity of optimal barycenters: there exists an optimal discrete barycenter whose support size is bounded by

$$K^* \leq \sum_{i=1}^n m_i - n + 1,$$

see the LP characterization and sparsity theorem of *Anderes, Borgwardt, and Miller* (2016, Theorems 1–2)[24]. Consequently, even though an optimal barycenter is sparse, the ambient LP may be very large because the candidate support set can grow combinatorially with $\{m_i\}$, which leads to substantial computational cost.

Following [25], we adopt the entropically regularized OT formulation to obtain a smooth and strictly convex approximation. For discrete measures $\mu = \sum_p a_p \delta_{x_p}$ and $\nu = \sum_q b_q \delta_{y_q}$ with ground cost $M_{pq} = \|x_p - y_q\|^2$, we consider

$$p_\varepsilon(a, b; M) = \min_{T \in U(a, b)} \langle T, M \rangle - \varepsilon H(T), \quad H(T) = - \sum_{p, q} T_{pq} \log T_{pq}, \quad (5)$$

whose dual involves potentials (f, g) and can be computed efficiently by Sinkhorn matrix scaling.

In the semi-free support setting (Algorithm 1), we fix a target support size $k \ll \sum_i m_i$ and jointly optimize the barycenter weights $a \in \Delta^k$ and locations $X = \{x_1, \dots, x_k\} \subset \mathbb{R}^d$. The algorithm alternates between (i) updating a using averaged dual potentials and (ii) updating X by weighted centroid aggregation induced by the soft transport maps. Our implementation follows the entropic smoothing approach of [25] while computing the dual potentials in the *log-domain* for numerical stability, which avoids underflow in the exponential scaling and enables stable large-scale computations.

Compared with the LP-based fully free-support approach [24], fixing k controls the problem size and yields a scalable approximation; compared with [25], the log-domain Sinkhorn implementation preserves the same smoothed barycenter objective while improving numerical robustness.

Algorithm 1 Free-support Wasserstein barycenter under fixed ε (normalized cost).

- 1: **Input:** a collection of point clouds $\{Y_i\}_{i=1}^n \subset \mathbb{R}^d$; number of support points k ; regularization parameter ε ; learning rate η ; weight mixing rate λ ; temperature T ; scale factor **scale2**.
 - 2: **Output:** barycenter supports $X \in \mathbb{R}^{k \times d}$ and weights $a \in \Delta^k$.
 - 3: Initialize $a = (1/k)\mathbf{1}_k$ and $X^{(0)}$ using k -means++ and FPS on the concatenated clouds.
 - 4: For each $i = 1, \dots, n$, compute initial dual potentials (f_i, g_i) via `Sinkhorn_log_domain`($X^{(0)}, Y_i, a, w_i, \varepsilon$).
 - 5: **repeat**
 - 6: $X^{\text{prev}} \leftarrow X, \quad a^{\text{prev}} \leftarrow a$
 - 7: $\bar{a} = -\varepsilon(1/n) \sum_{i=1}^n f_i$
 - 8: $a_{\text{new}} \propto \exp(-\bar{a}/T)$, normalize and mix uniformly
 - 9: $a \leftarrow (1 - \lambda)a + \lambda a_{\text{new}}$
 - 10: Compute $M_i = \text{row_mass_and_moment_softmax}(K_i, g_i, Y_i, a)$ for $i = 1, \dots, n$
 - 11: $X \leftarrow (1 - \eta)X + \eta(\sum_{i=1}^n M_i/n)/a$
 - 12: For each i , update (f_i, g_i) via `Sinkhorn_log_domain`($X, Y_i, a, w_i, \varepsilon, f_i, g_i$)
 - 13: $\delta_X = \|X - X^{\text{prev}}\|_2 / (\|X^{\text{prev}}\|_2 + 10^{-16})$
 - 14: $\delta_a = \|a - a^{\text{prev}}\|_1$
 - 15: **if** $\delta_X < \delta_0$ **and** $\delta_a < \delta_0$ **then**
 - 16: **return** (X, a) and stop
 - 17: **until** convergence
-

2.2 Fréchet Median in Wasserstein Space

The Fréchet median of a family of probability measures $\{\mu_i\}_{i=1}^N \subset \mathcal{P}_2(\mathbb{R}^d)$ is defined as the element that minimizes the sum of pairwise Wasserstein-2 distances to all members of the set:

$$\mu^* = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{i=1}^N W_2(\nu, \mu_i), \quad (6)$$

where $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of probability measures with finite second moment.

To compute μ^* , we adopt an *iteratively reweighted least squares* (IRLS) scheme defined on the Wasserstein space, which reformulates the median estimation problem as a sequence of weighted

Wasserstein barycenter computations. At each iteration t , the algorithm updates the weights according to the current distances between the candidate median $\nu^{(t)}$ and all input measures:

$$w_n^{(t)} = \frac{1}{W_2(\nu^{(t)}, \mu_n)}, \quad \tilde{w}_n^{(t)} = \frac{w_n^{(t)}}{\sum_{k=1}^N w_k^{(t)}}, \quad (7)$$

Given these normalized weights $\tilde{w}_n^{(t)}$, the next iterate $\nu^{(t+1)}$ is obtained by solving the corresponding weighted barycenter problem:

$$\nu^{(t+1)} = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{n=1}^N \tilde{w}_n^{(t)} W_2^2(\nu, \mu_n). \quad (8)$$

This update replaces the non smooth median objective with a locally quadratic surrogate, yielding a smooth majorization–minimization step in Wasserstein space.

In practice, each weighted barycenter subproblem (8) can be efficiently solved using the semi-free support barycenter method introduced in the previous subsection. By fixing a moderate number of support points and jointly optimizing their locations and weights under entropic regularization, we obtain a numerically stable and computationally tractable approximation of the Wasserstein median. The resulting estimator can thus be viewed as a semi-free support Wasserstein median, whose computation is summarized in Algorithm 2 [26].

Algorithm 2 Wasserstein median computation by IRLS.[26]

- 1: **Input:** a collection of probability measures $\{\mu_n\}_{n=1}^N$, weights $(\pi_1, \dots, \pi_N) \in \Delta_0^N$.
 - 2: **Output:** $\nu^* = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{n=1}^N \pi_n W_2(\nu, \mu_n)$.
 - 3: **Initialize** $\nu^{(0)} \in \mathcal{P}_2(\mathbb{R}^d)$.
 - 4:
 - 5: **repeat**
 - 6: $w_n^{(t)} = \pi_n / W_2(\nu^{(t)}, \mu_n)$
 - 7: $\tilde{w}_n^{(t)} = w_n^{(t)} / \sum_{k=1}^N w_k^{(t)}$
 - 8: $\nu^{(t+1)} = \arg \min_{\nu \in \mathcal{P}_2(\mathbb{R}^d)} \sum_{n=1}^N \tilde{w}_n^{(t)} W_2^2(\nu, \mu_n)$
 - 9: **if** $\nu^{(t+1)}$ equals one of the μ_n 's **or** $\nu^{(t+1)}$ is close to $\nu^{(t)}$ **then**
 - 10: **return** $\nu^{(t+1)}$ and stop
 - 11: **until** convergence
-

2.3 K-Median Clustering in Wasserstein Space

The optimization of the Wasserstein K -median objective is a highly nonconvex problem [27, 28, 25], since both cluster assignments and Wasserstein medians are jointly unknown and the median in Wasserstein space is itself not necessarily unique. Consequently, direct alternating minimization between assignments and cluster medians often suffers from sensitivity to initialization and can easily get trapped in poor local minima.

Formally, given empirical measures $\{\mu_i\}_{i=1}^n \subset \mathcal{P}_2(\Omega)$, the goal of Wasserstein K -median clustering is to find partitions $\{S_k\}_{k=1}^K$ and cluster representatives $\{m_k\}_{k=1}^K$ that minimize the total within-cluster Wasserstein distance:

$$\min_{\{S_k, m_k\}} \sum_{k=1}^K \sum_{i \in S_k} W_2(\mu_i, m_k). \quad (9)$$

Each cluster center m_k corresponds to a Wasserstein Fréchet median of its assigned members.

Simulated Annealing for Robust Optimization. To enhance global convergence and improve robustness, we adopt a *simulated annealing* (SA) strategy [29, 30, 31] for Wasserstein K -median

clustering. In this framework, the hard cluster assignments are replaced by probabilistic ones governed by a Gibbs distribution:

$$\Pr[c_i = k] = \frac{\exp(-W_2(\mu_i, m_k)/T)}{\sum_{\ell=1}^K \exp(-W_2(\mu_i, m_\ell)/T)}, \quad (10)$$

where $T > 0$ denotes the temperature. At high temperatures, assignments are nearly uniform, allowing wide exploration of the solution space; as T decreases, assignments become increasingly deterministic, leading to refined partitions.

At each iteration and temperature level, the algorithm alternates between: (i) updating the assignment probabilities according to (10), and (ii) recomputing each cluster center as a Wasserstein Fréchet median of the weighted subset:

$$m_k^{(t+1)} = \arg \min_{\nu \in \mathcal{P}_2(\Omega)} \sum_{i=1}^n p_{i,k}^{(t)} W_2^2(\nu, \mu_i), \quad (11)$$

where $p_{i,k}^{(t)} = \Pr[c_i = k]$. Each median update in (11) is computed using the semi-free support IRLS scheme introduced in the previous section [26]. After every full iteration, the temperature is geometrically cooled as

$$T \leftarrow \alpha T, \quad 0 < \alpha < 1. \quad (12)$$

This simulated annealing process provides a principled stochastic relaxation of the discrete clustering problem and empirically yields solutions closer to the global optimum compared with deterministic updates. The complete procedure is summarized in Algorithm ??.

Algorithm 3 Wasserstein K -median Clustering (Simulated Annealing)

- 1: **Input:** measures $\{\mu_1, \dots, \mu_n\} \subset \mathcal{P}_2(\Omega)$; cluster count K ; initial temperature T_0 ; cooling rate α ($0 < \alpha < 1$); tolerance ε
 - 2: **Initialize:** Randomly pick K indices $n_1, \dots, n_K$; set $m_k \leftarrow \mu_{n_k}$; $T \leftarrow T_0$
 - 3: **repeat**
 - 4: **for** $i \leftarrow 1$ **to** n **do**
 - 5: **for** $k \leftarrow 1$ **to** K **do**
 - 6: $d_{i,k} \leftarrow W_2(\mu_i, m_k)$
 - 7: Draw label c_i from $\Pr[c_i = k] = \frac{\exp(-d_{i,k}/T)}{\sum_{k=1}^K \exp(-d_{i,k}/T)}$
 - 8: **for** $k \leftarrow 1$ **to** K **do**
 - 9: $S_k \leftarrow \{i : c_i = k\}$
 - 10: $m_k^{\text{new}} \leftarrow \text{WASSERSTEIN MEDIAN}(\{\mu_i\}_{i \in S_k})$
 - 11: $\Delta \leftarrow \max_k W_2(m_k^{\text{new}}, m_k)$
 - 12: $m_k \leftarrow m_k^{\text{new}} \quad (k = 1, \dots, K)$
 - 13: $T \leftarrow \alpha T$
 - 14: **until** $\Delta \leq \varepsilon$
 - 15: **return** labels $(c_1, \dots, c_n)$, Wasserstein medians $(m_1, \dots, m_K)$
-

PAM-based Warm Start. While the simulated annealing strategy offers global robustness, its stochastic nature and repeated median computations can be computationally demanding for large datasets. To accelerate convergence and reduce sensitivity to random initialization, we employ the Partitioning Around Medoids (PAM) heuristic [32] as a deterministic warm-start procedure that leverages the pairwise Wasserstein geodesic distances between distributions.

Formally, given empirical measures $\{\mu_i\}_{i=1}^n \subset \mathcal{P}_2(\Omega)$, we first compute the pairwise Wasserstein distance matrix

$$D_{ij} = W_2(\mu_i, \mu_j), \quad i, j = 1, \dots, n. \quad (13)$$

The PAM heuristic is then applied to identify K representative medoids that minimize the total within-cluster dispersion:

$$\min_{\{S_k, m_k\}} \sum_{k=1}^K \sum_{i \in S_k} D_{i, m_k}. \quad (14)$$

The resulting medoids $\{m_k^{(0)}\}_{k=1}^K$ and initial cluster assignments $\{c_i^{(0)}\}_{i=1}^n$ provide a good initialization for the subsequent Wasserstein K -median optimization.

Starting from this initialization, we perform deterministic refinement through standard assignment and median-update iterations: each measure μ_i is reassigned to the nearest current center, and each cluster center m_k is updated as the Wasserstein median of its members:

$$m_k^{(1)} = \arg \min_{\nu \in \mathcal{P}_2(\Omega)} \sum_{i \in S_k^{(t)}} W_2^2(\nu, \mu_i). \quad (15)$$

This two-stage approach—PAM initialization followed by iterative refinement—achieves fast convergence and improved stability, while retaining the interpretability and geometric consistency of the Wasserstein K -median formulation. The overall procedure is summarized in Algorithm ??.

Algorithm 4 Wasserstein K -median Clustering with PAM Warm Start

```

1: Input: measures  $\{\mu_1, \dots, \mu_n\} \subset \mathcal{P}_2(\Omega)$ ; cluster count  $K$ ; tolerance  $\varepsilon$ 
2: PAM warm start:
   Compute pairwise distances  $D_{ij} = W_2(\mu_i, \mu_j)$ 
   Run PAM on  $D$  to obtain medoid indices  $n_1, \dots, n_K$  and initial labels  $\{c_i^{(0)}\}$ 
   Set  $m_k \leftarrow \mu_{n_k}$  for  $k = 1, \dots, K$ 
3: repeat
4:   for  $i \leftarrow 1$  to  $n$  do
5:      $c_i \leftarrow \arg \min_k W_2(\mu_i, m_k)$ 
6:   for  $k \leftarrow 1$  to  $K$  do
7:      $S_k \leftarrow \{i : c_i = k\}$ 
8:      $m_k^{\text{new}} \leftarrow \text{WASSERSTEIN MEDIAN}(\{\mu_i\}_{i \in S_k})$ 
9:    $\Delta \leftarrow \max_k W_2(m_k^{\text{new}}, m_k)$ 
10:   $m_k \leftarrow m_k^{\text{new}}$  ( $k = 1, \dots, K$ )
11: until  $\Delta \leq \varepsilon$ 
12: return labels  $(c_1, \dots, c_n)$ , medians  $(m_1, \dots, m_K)$ 

```

2.4 Validation by Wasserstein Spatial Depth

Data depth provides a center-outward ordering of multivariate or functional observations, serving as a robust measure of centrality and outlyingness in a statistical population [33, 22]. Unlike distance-based criteria, depth functions are affine-invariant and resistant to outliers, making them particularly suitable for analyzing nonconvex or heterogeneous structures.

In Wasserstein space, the notion of depth can be generalized to distributional data. Following [2], the *Wasserstein spatial depth* (WSD) of a distribution $Q \in \mathcal{P}_2^{a.c}(\mathbb{R}^d)$ with respect to a distribution of distributions $\mathbb{P} \in \mathcal{P}(\mathcal{P}_2(\mathbb{R}^d))$ is defined as

Definition 2.1 (Wasserstein spatial depth, [2]).

$$\text{WSD}(Q; \mathbb{P}) := 1 - \left(\int \left\| \mathbb{E}_{P \sim \mathbb{P}} \left[\frac{x - T_{Q,P}(x)}{W_2(P, Q)} \right] \right\|^2 dQ(x) \right)^{1/2}.$$

When $\mathbb{P}_{P \sim \mathbb{P}}(W_2(P, Q) = 0) \neq 0$, we set $\frac{x - T_{Q,P}(x)}{W_2(P, Q)} = 0$ for such cases.

Building on the general concept of data depth, Jörnsten, Vardi, and Zhang [23] introduced the idea of *relative depth* (ReD) as a robust internal validation index for clustering. ReD compares within- and between-cluster depths, emphasizing centrality and separation rather than distance or variance. Unlike the silhouette score, ReD is invariant to scale and remains reliable in noisy or high-dimensional contexts.

We extend this idea to Wasserstein space and define the *Wasserstein relative depth* (W-ReD) to evaluate clustering consistency and separability. For a collection of clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ and a distributional observation Q , let $D(Q; \mathcal{C}_k)$ denote its Wasserstein spatial depth with respect to cluster $\mathcal{C}_k$. The

relative depth of Q is defined as

$$\text{ReD}(Q) = D(Q; \mathcal{C}_{k^*}) - \max_{j \neq k^*} D(Q; \mathcal{C}_j), \quad k^* = \arg \max_k D(Q; \mathcal{C}_k). \quad (16)$$

A higher $\text{ReD}(Q)$ indicates that Q is considerably deeper within its own cluster than in others, implying a confident assignment.

In practice, $D(Q; \mathcal{C}_k)$ is computed via the WSD, allowing ReD to capture cluster centrality directly in Wasserstein space. This framework generalizes the classical ReD of Jörnsten, Vardi, and Zhang [23] to distribution-valued data, yielding a robust, geometry-aware alternative to distance-based validation measures such as the silhouette score.

Finally, the optimal number of clusters K^* can be determined by maximizing the average relative depth across all observations:

$$K^* = \arg \max_K \frac{1}{N} \sum_{i=1}^N \text{ReD}_K(Q_i), \quad (17)$$

where $\text{ReD}_K(Q_i)$ denotes the relative depth of Q_i under the clustering solution with K clusters. This criterion favors partitions that are both internally cohesive and externally well separated in terms of Wasserstein centrality.

3 Theoretical Guarantee

3.1 Consistency of K-means in Wasserstein Space

Let (X, d) be a Polish metric space and let $\mathcal{P}_2(X)$ denote the space of Borel probability measures on X with finite second moment, endowed with the 2-Wasserstein distance

$$W_2(\mu, \nu) := \left(\inf_{\pi \in \Pi(\mu, \nu)} \int d(x, y)^2 d\pi(x, y) \right)^{1/2}, \quad (18)$$

where $\Pi(\mu, \nu)$ denotes the set of all couplings of μ and ν .

We observe i.i.d. random probability measures

$$\mu_1, \mu_2, \dots \sim \mathbb{P},$$

with empirical distribution

$$\mathbb{P}_n := \frac{1}{n} \sum_{i=1}^n \delta_{\mu_i}.$$

Given a finite center set $A \subset \mathcal{P}_2(X)$, the Wasserstein k -means functional is defined by [34]

$$\Phi(A, Q) := \int \min_{\nu \in A} W_2(\mu, \nu)^2 Q(d\mu). \quad (19)$$

The corresponding population and empirical optimal values are

$$m_k(Q) := \inf \{ \Phi(A, Q) : |A| \leq k \}, \quad \Phi(A_n(k), \mathbb{P}_n) = m_k(\mathbb{P}_n),$$

and a population minimizer is denoted

$$\bar{A}(k) \in \arg \min_{|A| \leq k} \Phi(A, \mathbb{P}).$$

Under compactness of the base space (X, d) , the Wasserstein space $\mathcal{P}_2(X)$ is itself compact under W_2 ; hence both population and empirical minimizers exist. This provides a natural analogue of the Euclidean k -means setting studied by Pollard, and allows the full Pollard-style consistency machinery to be extended to Wasserstein space. In particular, the objective $\Phi(A, Q)$ remains continuous in the Hausdorff metric on finite center sets, and the parameter space $\mathcal{A}_k := \{A \subset \mathcal{P}_2(X) : |A| \leq k\}$ is compact. These structural properties ensure that the empirical minimizers $A_n(k)$ are well-behaved and admit subsequential limits, enabling a direct adaptation of Pollard's three-step argument: (i) at least one empirical center cannot escape to infinity, (ii) in fact all centers remain in a common compact region, and (iii) uniform convergence of the empirical objective identifies this region's unique population minimizer. We now state the resulting consistency theorem.

3.1.1 Main Theorem

Theorem 3.1 (Consistency of Wasserstein k -Means). *Assume:*

- (i) (X, d) is a compact metric space. Hence $\mathcal{P}_2(X)$ is compact under the W_2 metric.
- (ii) For some (and therefore every) $\mu_0 \in \mathcal{P}_2(X)$,

$$\int_{\mathcal{P}_2(X)} W_2(\mu, \mu_0)^2 \mathbb{P}(d\mu) < \infty.$$

- (iii) For each $j = 1, \dots, k$, the population minimizer $\bar{A}(j)$ satisfying

$$\Phi(\bar{A}(j), \mathbb{P}) = m_j(\mathbb{P})$$

is unique up to permutation.

Then the empirical k -means centers are strongly consistent:

$$A_n(k) \longrightarrow \bar{A}(k) \quad \text{a.s. in the Hausdorff metric,}$$

and the empirical optimal values satisfy

$$\Phi(A_n(k), \mathbb{P}_n) \rightarrow m_k(\mathbb{P}) \quad \text{a.s.}$$

3.1.2 Proof of Theorem 3.1

For a finite set $A \subset \mathcal{P}_2(X)$,

$$\Phi(A, Q) = \int \min_{\nu \in A} W_2(\mu, \nu)^2 Q(d\mu), \quad m_k(Q) = \inf\{\Phi(A, Q) : |A| \leq k\}.$$

Compactness of $\mathcal{P}_2(X)$ ensures existence of both $\bar{A}(k)$ and $A_n(k)$.

Step 1: At least one empirical center lies in a fixed ball. Fix $\mu_0 \in \mathcal{P}_2(X)$. Since $\mathbb{P}$ is supported on a compact set, there exists $r > 0$ such that $K = B_{W_2}(\mu_0, r)$ satisfies $\mathbb{P}(K) > 0$.

Assume, for contradiction, that $A_n(k)$ contains no point inside $B_{W_2}(\mu_0, M)$ for infinitely many n . Then for any $\mu \in K$ and $\nu \in A_n(k)$,

$$W_2(\mu, \nu) \geq M - r,$$

and therefore

$$\Phi(A_n(k), \mathbb{P}_n) \geq (M - r)^2 \mathbb{P}_n(K).$$

Taking $\liminf$ gives

$$\liminf_n \Phi(A_n(k), \mathbb{P}_n) \geq (M - r)^2 \mathbb{P}(K).$$

Choose M so that

$$(M - r)^2 \mathbb{P}(K) > \Phi(\{\mu_0\}, \mathbb{P}) = \int W_2(\mu, \mu_0)^2 \mathbb{P}(d\mu),$$

which contradicts optimality of $A_n(k)$. Thus $A_n(k) \cap B_{W_2}(\mu_0, M) \neq \emptyset$ eventually.

Step 2: All empirical centers lie in a larger ball. Suppose there exist infinitely many n and $\nu_n \in A_n(k)$ for which

$$W_2(\nu_n, \mu_0) > 5M.$$

Step 1 ensures that $A_n(k)$ also contains a point η_n with $W_2(\eta_n, \mu_0) \leq M$.

If $W_2(\mu, \mu_0) \leq 2M$, then

$$W_2(\mu, \nu_n) \geq 3M.$$

Hence the center ν_n does not influence assignment for such μ . Let $A'_n = A_n(k) \setminus \{\nu_n\}$.

Choose M large enough so that

$$\int_{\{W_2(\mu, \mu_0) \geq 2M\}} (5M - W_2(\mu, \mu_0))_+^2 \mathbb{P}(d\mu) < m_k(\mathbb{P}) - m_{k-1}(\mathbb{P}),$$

which forces, for large n ,

$$\Phi(A_n(k), \mathbb{P}_n) - \Phi(A'_n, \mathbb{P}_n) > m_k(\mathbb{P}) - m_{k-1}(\mathbb{P}),$$

contradicting optimality of $A_n(k)$. Thus $A_n(k) \subset B_{W_2}(\mu_0, 5M)$ eventually.

Step 3: Identification of the limit. Since $A_n(k)$ is eventually contained in the compact set $B_{W_2}(\mu_0, 5M)$, the family

$$\mathcal{A}_k := \{A \subset B_{W_2}(\mu_0, 5M) : |A| \leq k\}$$

is compact under the Hausdorff metric. Hence any subsequence admits a Hausdorff-convergent refinement $A_{n_j}(k) \rightarrow A^*$. To identify the limit A^* , two ingredients are required: continuity of $A \mapsto \Phi(A, \mathbb{P})$ on $\mathcal{A}_k$ and a uniform law of large numbers for the associated function class.

For $A \subset \mathcal{P}_2(X)$ finite write

$$g_A(\mu) = \min_{\nu \in A} W_2(\mu, \nu)^2.$$

If $H(A, B) < \delta$ then for each $b \in B$ there exists $a(b) \in A$ with $W_2(a(b), b) < \delta$. By the triangle inequality,

$$W_2(\mu, b) \leq W_2(\mu, a(b)) + W_2(a(b), b),$$

hence

$$g_B(\mu) = \min_{b \in B} W_2(\mu, b)^2 \leq \min_{b \in B} (W_2(\mu, a(b)) + \delta)^2.$$

Expanding the square and using boundedness of the parameter space,

$$g_B(\mu) - g_A(\mu) \leq C \delta$$

for some constant C depending only on the diameter of $B_{W_2}(\mu_0, 5M)$. Integrating against $\mathbb{P}$ gives

$$|\Phi(A, \mathbb{P}) - \Phi(B, \mathbb{P})| \leq C \delta,$$

so $A \mapsto \Phi(A, \mathbb{P})$ is continuous on $\mathcal{A}_k$.

Define

$$\mathcal{F}_k = \{g_A : \mu \mapsto \min_{\nu \in A} W_2(\mu, \nu)^2 : A \in \mathcal{A}_k\}.$$

To establish a uniform SLLN for $\mathcal{F}_k$, we follow the approximation scheme of Gänssler and Stute used by Pollard. Fix $\varepsilon > 0$. Compactness of $\mathcal{A}_k$ ensures the existence of a finite δ -net $\mathcal{A}_{k,\delta}$ such that for every $A \in \mathcal{A}_k$ there exists $A' \in \mathcal{A}_{k,\delta}$ with $H(A, A') < \delta$. The continuity result above implies

$$\|g_A - g_{A'}\|_{L^1(\mathbb{P})} < \varepsilon,$$

provided δ is chosen small enough. Since $\mathcal{A}_{k,\delta}$ is finite, the class $\{g_{A'} : A' \in \mathcal{A}_{k,\delta}\}$ satisfies the (ordinary) SLLN, and the standard bracketing argument yields

$$\sup_{A \in \mathcal{A}_k} |\Phi(A, \mathbb{P}_n) - \Phi(A, \mathbb{P})| \rightarrow 0 \quad \text{a.s.}$$

For the subsequence $A_{n_j}(k)$,

$$\Phi(A_{n_j}(k), \mathbb{P}_{n_j}) \rightarrow \Phi(A^*, \mathbb{P}),$$

by the uniform SLLN, while optimality gives

$$\Phi(A_{n_j}(k), \mathbb{P}_{n_j}) = m_k(\mathbb{P}_{n_j}) \rightarrow m_k(\mathbb{P}).$$

Hence $\Phi(A^*, \mathbb{P}) = m_k(\mathbb{P})$, and the uniqueness assumption forces $A^* = \bar{A}(k)$. Since every subsequence converges to $\bar{A}(k)$, the full sequence does as well.

4 Numerical Results

4.1 Clustering and Validation on Sparse and Compact Distribution Families

To further assess the stability of the proposed Wasserstein depth-based validation method, we designed a simulation involving two Gaussian distribution families with drastically different spatial scales. As shown in Figure 1, Family A consists of Gaussian point-clouds whose means are uniformly distributed inside a large disk of radius $R_A = 100$ centered at $(0, 0)$, while Family B contains compact Gaussian clouds with means sampled within a small disk of radius $R_B = 0.1$ centered at $(100, 0)$. Each cloud represents an empirical measure $\mu_i \in \mathcal{P}_2(\mathbb{R}^2)$ with isotropic covariance $\Sigma = I_2$. This configuration yields a highly heterogeneous dataset where one cluster (Family A) spans a broad, sparse region, whereas the other (Family B) is extremely concentrated.

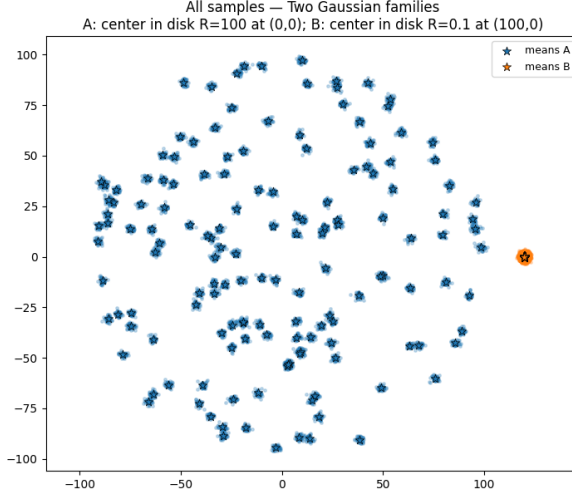


Figure 1: Two-family simulation setup: Family A (blue) forms a wide, sparse Gaussian cloud over a large disk, while Family B (orange) forms a tight, compact cluster in a small disk offset at $(100, 0)$. Stars mark the true mean centers of both families.

We applied Wasserstein K -means clustering to this dataset, using the squared $\mathcal{W}_2$ distance as the dissimilarity metric, and evaluated the resulting partitions via the classical silhouette score, the Davies–Bouldin Index (DBI), and the proposed Wasserstein relative depth (ReD). The numerical results are summarized in Table 4.1.

K	Silhouette	ReD (mean)	DBI
2	0.2730	0.2721	1.3547
3	0.3147	0.3032	1.0162
4	0.3319	0.2571	0.9997
5	0.3429	0.2740	0.9045
6	0.3387	0.2798	0.8784
Best K	5	3	6

Distance-based validation indices show a systematic tendency toward larger values of K in this heterogeneous setting. The silhouette score reaches its maximum at $K = 5$, while DBI attains its minimum at $K = 6$, both reflecting the same underlying issue: because these indices rely on global dispersion measures and average pairwise distances, they favor splitting the broad, low-density Family A into smaller, artificially compact subclusters. As a result, both metrics substantially overestimate the true number of clusters.

By contrast, the ReD statistic reaches its maximum at $K = 3$, which is much closer to the true underlying two-cluster generative structure. ReD measures each distribution’s relative centrality within

its assigned cluster rather than relying on global compactness, allowing it to remain stable under severe differences in density and spatial scale. This robustness highlights an important advantage of Wasserstein depth-based validation: it provides consistent and interpretable assessment across heterogeneous distribution families without being misled by variations in spread or point density.

4.2 Clustering Under Imbalanced Location and Scale Families

To further assess the behavior of different internal validation indices in heterogeneous distributional settings, we constructed two simulation scenarios on $\mathcal{P}_2(\mathbb{R}^2)$ that intentionally create strong imbalances: one based on *location separation* and the other based on *scale variation*. Both examples illustrate that Wasserstein geometry captures not only positional differences between distributions but also differences in dispersion and shape.

(A) Location-imbalanced Gaussian Families

In the first simulation, we generated three Gaussian families with identical within-cloud variance but highly imbalanced spatial locations. For families 1 and 2, Gaussian means were drawn from two nearby uniform bands and arranged as horizontal mirror images:

$$\begin{aligned} x_1 &\sim \text{Uniform}[-1.8, -1.7], & y_1 &\sim \text{Uniform}[0, 0.5], \\ x_2 &\sim \text{Uniform}[-0.3, -0.2], & y_2 &\sim \text{Uniform}[0, 0.5]. \end{aligned}$$

The third family was placed far apart:

$$x_3 \sim \text{Uniform}[20, 21], \quad y_3 \sim \text{Uniform}[0, 0.5].$$

Each empirical measure was generated from

$$X_i^{(k)} \sim \mathcal{N}(\mu_i^{(k)}, \sigma^2 I_2), \quad \sigma = 0.12, \quad m = 50.$$

This produces 600 empirical distributions: two tightly packed clusters near the origin and one isolated cluster roughly 20 units away (Figure 2).

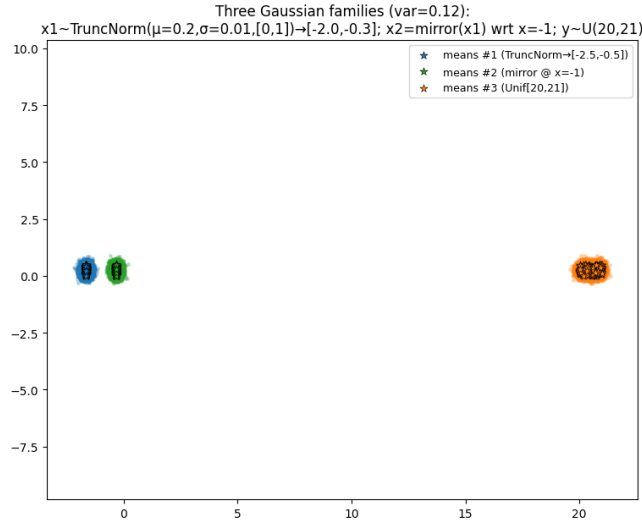


Figure 2: Location-imbalanced Gaussian families. Families 1 lies near $x \approx [-1.8, -1.7]$, Families 2 lies near $x \approx [-0.3, -0.2]$, while family 3 is placed at $x \approx [20, 21]$. Each point cloud is one empirical measure.

We applied Wasserstein K -median clustering with $K \in \{2, 3, 4, 5, 6\}$. Representative results are:

K	Silhouette	ReD (mean)	DBI
2	0.9670	0.3986	0.0501
3	0.9123	0.4003	0.1285
4	0.8036	0.3762	0.3521
5	0.6419	0.3504	0.5783
6	0.6047	0.3356	0.7125
Best K	2	3	2

Both silhouette and DBI favor $K = 2$, merging the two nearby families because the extremely distant third component dominates pairwise separation. In contrast, ReD correctly identifies $K = 3$, since depth-based centrality remains sensitive to local structure even under extreme inter-cluster distance imbalance.

(B) Scale-imbalanced Gaussian Families

In the second simulation, we tested the ability of validation indices to detect clusters that differ purely by *scale*. All empirical measures were centered at the same location $(0, 0)$, but had different variances sampled from triangular distributions:

$$\sigma_A \sim \text{Triangular}(0.0, 0.2, 0.4), \quad \sigma_B \sim \text{Triangular}(1.0, 1.2, 1.4), \quad \sigma_C \sim \text{Triangular}(5.0, 5.2, 5.4).$$

Each cloud was generated as

$$X_i^{(k)} \sim \mathcal{N}(0, \sigma_k^2 I_2), \quad m = 50,$$

producing $n_A = 50$, $n_B = 150$, and $n_C = 200$ empirical measures (Figure 3).

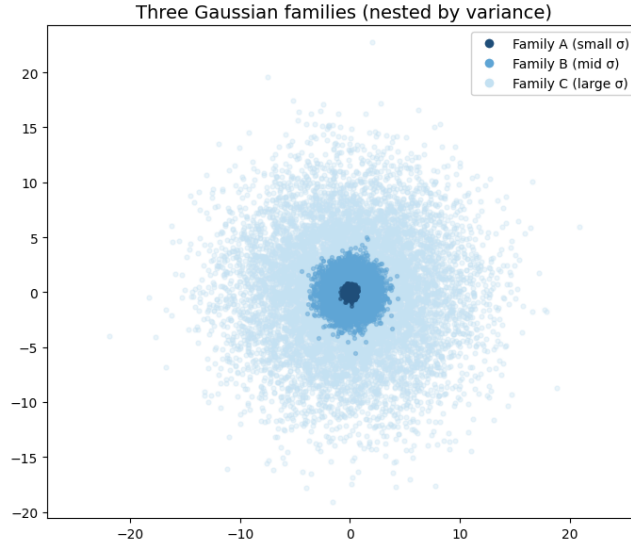


Figure 3: Scale-imbalanced families with identical means but variances drawn from different triangular distributions. Colors correspond to $(\sigma_A, \sigma_B, \sigma_C)$ groups.

Validation results are:

K	Silhouette	ReD (mean)	DBI
2	0.6623	0.3599	0.5810
3	0.5132	0.3649	0.5602
4	0.2938	0.1635	1.7536
5	0.1440	0.0715	2.3063
Best K	2	3	3

Silhouette again merges two families and selects $K = 2$, because all clusters share the same center and the smallest-variance group forms a compact core. DBI improves but still struggles due to high within-cluster variance imbalance. ReD successfully identifies $K = 3$, since Wasserstein depth directly reflects how the dispersion of each cloud positions it relative to the cluster center.

Summary

Across both location- and scale-imbalanced settings, ReD consistently recovers the true number of generating families, whereas silhouette and DBI systematically favor smaller K values. These experiments highlight an important geometric property of optimal transport: *Wasserstein distance captures not only positional differences but also shape- and scale-based variation*, enabling depth-based validation to remain stable and discriminative in heterogeneous distributional scenarios.

4.3 Clustering and Validation on a Noisy Distribution Family

To evaluate the performance of the proposed Wasserstein K -median clustering and the depth-based validation criterion, we designed a controlled simulation involving three Gaussian distribution families with added background noise. The three main families were placed approximately at the vertices of an equilateral triangle with side length about 2ρ . Let $c_1, c_2, c_3 \in \mathbb{R}^2$ denote these vertices, with

$$c_1 = (-\rho, 0), \quad c_2 = (\rho, 0), \quad c_3 = (0, \sqrt{3}\rho),$$

and set $\rho = 0.8$. For each family $k \in \{1, 2, 3\}$ we sampled $n_{\text{each}} = 60$ cloud centers $\mu_i^{(k)} \sim \mathcal{N}(c_k, \sigma_{\text{mean}}^2 I_2)$ with $\sigma_{\text{mean}} = 0.6$, and then generated point clouds

$$X_i^{(k)} \sim \mathcal{N}(\mu_i^{(k)}, \sigma_{\text{inner}}^2 I_2), \quad \sigma_{\text{inner}} = 0.08,$$

each consisting of $m = 100$ points in $\mathbb{R}^2$. Every cloud $X_i^{(k)}$ was interpreted as an empirical measure $\mu_i \in \mathcal{P}_2(\mathbb{R}^2)$.

To introduce structured noise, we first computed the axis-aligned bounding box of all Gaussian centers and enlarged it by a margin of size `box_pad` = 0.8. For each of the three families we then sampled 40 additional “noise centers” uniformly from this rectangle and generated Gaussian clouds with the same inner variance $\sigma_{\text{inner}}^2 I_2$. These noise clouds do not belong to any of the three main families, producing substantial overlap and boundary ambiguity in the resulting collection of $3n_{\text{each}} + 3 \times 40 = 300$ empirical distributions.

We applied the proposed Wasserstein K -median algorithm with the semi-free support IRLS median solver to this dataset. For each candidate number of clusters $K \in \{2, 3, 4, 5, 6\}$, we performed iterative assignment–median updates until convergence and recorded the *silhouette score*, the *Davies–Bouldin Index* (DBI), and the *mean relative depth* (ReD). The results are:

K	Silhouette	ReD (mean)	DBI
2	0.3224	0.2808	1.3234
3	0.3687	0.3101	0.9698
4	0.3796	0.3055	0.9491
5	0.3171	0.2984	0.9852
6	0.3449	0.2963	0.9327
Best K	4	3	6

Distance-based validation indices display a clear tendency toward larger values of K in this noisy, partially overlapping setting. The silhouette score reaches its maximum at $K = 4$, while DBI attains its minimum at $K = 6$. Both behaviors reflect their reliance on global dispersion and average pairwise distances: local density variation and background noise are accommodated by subdividing the true components into smaller, artificially compact subclusters, leading to an overestimation of the number of clusters.

In contrast, the ReD statistic achieves its maximum at $K = 3$, matching the true number of generating families. Because ReD is based on Wasserstein spatial depth, it quantifies each distribution’s relative centrality within its assigned cluster rather than boundary separation, and is therefore

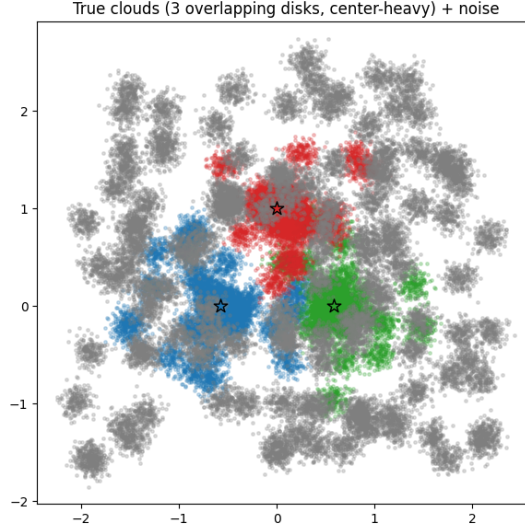


Figure 4: True simulated distributions for the triangle-plus-noise experiment. Colored point clouds correspond to the three Gaussian families located near the vertices of an equilateral triangle; gray clouds indicate uniformly distributed noise within the surrounding bounding box. Stars mark the triangle vertex centers $\{c_1, c_2, c_3\}$.

more robust to overlapping regions and peripheral noise [23, 22]. This yields a smoother and more interpretable assessment of cluster quality for complex distribution-valued data with structured noise.

4.4 Clustering and Validation on Heavy-Tailed Distribution Families

To examine the robustness of the proposed validation approach under heavy-tailed and heterogeneous settings, we construct a synthetic family of distribution-valued data whose cluster centers follow a Pareto-type radial distribution on $\mathcal{P}_2(\mathbb{R}^2)$. Three groups of empirical Gaussian clouds are placed near the vertices of an approximately equilateral triangle with side length about 2ρ . Let

$$c_1 = (-\rho, 0), \quad c_2 = (\rho, 0), \quad c_3 = (0, \sqrt{3}\rho),$$

with $\rho = 1.0$. For each family $k \in \{1, 2, 3\}$ we first draw $n_{\text{each}} = 120$ mean locations according to a Pareto radial law around c_k :

$$r \sim \text{Pareto}(r_{\min}, \alpha), \quad \theta \sim \text{Uniform}[0, 2\pi),$$

where $(x, y) = c_k + (r \cos \theta, r \sin \theta)$ gives the center of each local Gaussian component. We use $r_{\min} = 0.35$, a heavy-tail index $\alpha = 0.45$, and cap the radius at $r_{\max} = r_{\text{cap}} = 5.0$ to avoid extremely distant outliers. Smaller α induces heavier tails, producing dense central cores together with widely dispersed peripheral clouds. Given these means, we generate empirical Gaussian clouds

$$X_i^{(k)} \sim \mathcal{N}(\mu_i^{(k)}, \sigma^2 I_2), \quad \sigma = 0.2,$$

each containing $m = 50$ points in $\mathbb{R}^2$, and interpret every cloud $X_i^{(k)}$ as an empirical measure $\mu_i \in \mathcal{P}_2(\mathbb{R}^2)$. The resulting dataset combines three structured clusters with heavy-tailed radial spread and numerous peripheral clouds, as illustrated in Figure 5.

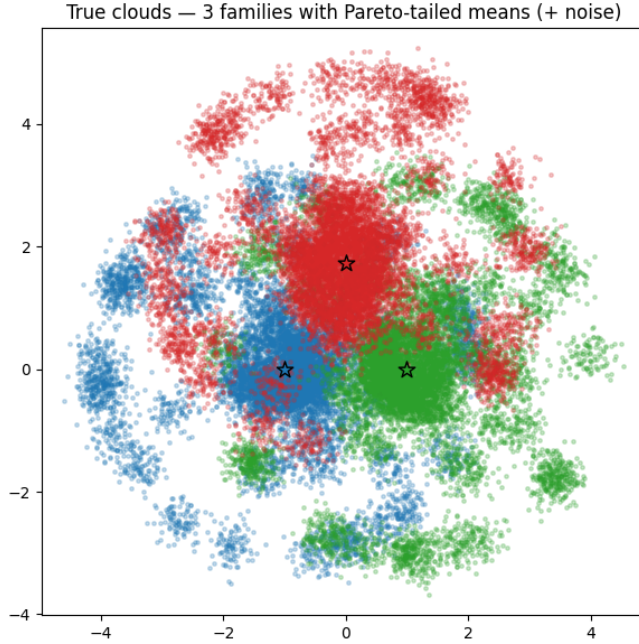


Figure 5: Visualization of the Pareto heavy-tailed distribution families. Three overlapping clusters (colored) exhibit center-heavy structures with Pareto-distributed means and dispersed peripheral clouds.

We apply the Wasserstein K -median clustering algorithm with PAM-based initialization to this dataset and evaluate the clustering performance using three validation metrics: the classical silhouette score, the Davies–Bouldin Index (DBI), and the proposed Wasserstein relative depth (ReD). The quantitative results across varying K values are summarized in Table 4.4.

K	Silhouette	ReD (mean)	DBI
2	0.2739	0.2731	1.7223
3	0.3163	0.3040	1.2903
4	0.2734	0.2761	1.1436
5	0.3094	0.2791	1.0057
6	0.3358	0.2786	0.9417
Best K	6	3	6

Distance-based validation indices show a clear tendency to select larger values of K in this heavy-tailed scenario. Both the silhouette score and DBI attain their optima at $K = 6$, substantially overestimating the true number of underlying clusters ($K = 3$). Because these indices are built from global dispersion and average pairwise distances, heavily tailed and peripheral clouds are accommodated by splitting the main components into smaller, artificially compact subclusters [13, 12].

In contrast, the ReD statistic reaches its maximum at $K = 3$, matching the true generating structure. ReD quantifies relative centrality in Wasserstein space rather than boundary separation, and is therefore less sensitive to heavy-tailed peripheral clouds and heterogeneous within-cluster spread [23, 22]. Consequently, ReD provides a more faithful summary of the intrinsic cluster structure in heterogeneous and heavy-tailed distributional data.

5 Real Data Analysis

5.1 MNIST Digits

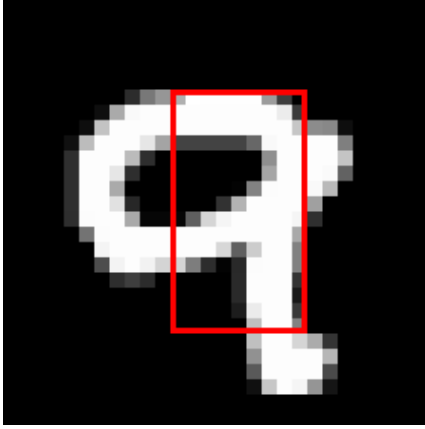
To illustrate the construction of distribution-valued data and evaluate the performance of the proposed Wasserstein depth-based validation framework on real-world data, we applied it to the MNIST

handwritten digit dataset. Each grayscale image of size 28×28 was treated as a discrete empirical measure over pixel coordinates, where pixel intensities were normalized to sum to one:

$$\mu_i = \sum_{p=1}^{784} w_{ip} \delta_{x_p}, \quad w_{ip} = \frac{I_{ip}}{\sum_{q=1}^{784} I_{iq}},$$

with $x_p \in [0, 1]^2$ denoting the normalized pixel position and I_{ip} the grayscale intensity at pixel p in image i . This transformation embeds each image as a probability distribution $\mu_i \in \mathcal{P}_2([0, 1]^2)$, enabling clustering directly in Wasserstein space.

Figure 6 provides an intuitive example of this transformation. The left panel shows a raw MNIST image, with a 16×9 subregion highlighted in red. The corresponding normalized weight submatrix is displayed on the right, demonstrating how grayscale pixel intensities are converted into probability weights that together form an empirical distribution for each image.



(a) Raw MNIST image with 16×9 subregion (red box).

$\Rightarrow$

0.00434	0.00628	0.00626	0.00626	0.00626	0.00626	0.00384	0.00153	0
0.00623	0.00623	0.00623	0.00623	0.00623	0.00623	0.00623	0.00581	0.00143
0.00623	0.00623	0.00623	0.00623	0.00623	0.00623	0.00623	0.00597	0.00492
0.00623	0.00623	0.00623	0.00623	0.00623	0.00623	0.00592	0.00433	0.00261
0.00623	0.00623	0.00623	0.00623	0.00623	0.00599	0.00420	0.00150	0.00018
0.00623	0.00623	0.00623	0.00623	0.00623	0.00552	0.00302	0.00061	0
0.00623	0.00623	0.00623	0.00623	0.00553	0.00322	0.00102	0.00013	0
0.00623	0.00623	0.00623	0.00578	0.00381	0.00159	0.00024	0	0
0.00623	0.00623	0.00563	0.00348	0.00133	0.00021	0	0	0
0.00623	0.00591	0.00424	0.00179	0.00034	0	0	0	0
0.00618	0.00460	0.00190	0.00037	0	0	0	0	0
0.00487	0.00227	0.00044	0	0	0	0	0	0
0.00260	0.00063	0.00010	0	0	0	0	0	0
0.00127	0.00017	0	0	0	0	0	0	0
0.000714	0.00579	0.00623	0.00623	0.00623	0.00623	0.00623	0.00623	0.00140
0	0.00473	0.00623	0.00623	0.00623	0.00623	0.00623	0.00623	0.00318

(b) Normalized weight submatrix corresponding to the red-box region.

Figure 6: Local region visualization of MNIST as an empirical distribution. The right-hand matrix displays the globally normalized pixel intensities restricted to the 16×9 red-box region (left), illustrating how image intensities are transformed into probability weights for Wasserstein analysis.

We selected digits labeled as “1”, “2”, and “9”, taking 30 images from each class to form a total of $N = 90$ normalized point-cloud distributions. The Wasserstein K -median clustering algorithm was applied using barycenter solver with entropic regularization. For each $K \in \{2, \dots, 9\}$, we computed both the silhouette score and the mean Wasserstein relative depth (ReD) to assess clustering validity.

K	Silhouette	ReD (mean)
2	0.2429	0.1453
3	0.2794	0.1730
4	0.2754	0.1712
5	0.2541	0.1533
6	0.2872	0.1576
7	0.2601	0.1450
8	0.2447	0.1428
9	0.2176	0.1247
Best K	6	3

Both indices show decreasing trends as K increases, but their optima differ notably. The silhouette score peaks at $K = 6$, indicating a tendency to over-segment each digit class into smaller subgroups due to shape variation and intra-class heterogeneity [12, 13]. In contrast, the Wasserstein relative depth (ReD) achieves its maximum at $K = 3$, which exactly corresponds to the true number of digit

families. Because ReD measures relative centrality rather than distance-based separation, it remains stable under morphological variability and illumination differences between digits. This experiment demonstrates that depth-based validation provides a robust and interpretable criterion for estimating the number of clusters in real distribution-valued data, highlighting its advantage in identifying meaningful structure in Wasserstein space.

5.2 Acute Myeloid Leukemia (AML) Flow Cytometry Cohort: Clustering and Validation

To further assess the applicability of the proposed Wasserstein K -median clustering and depth-based validation framework, we applied our method to a real flow cytometry dataset derived from an acute myeloid leukemia (AML) cohort. Flow cytometry quantifies fluorescence-labeled protein markers at the single-cell level and remains a central tool for characterizing hematopoietic populations and leukemic phenotypes [36]. AML bone-marrow samples are known to exhibit substantial immunophenotypic heterogeneity across patients, reflecting dysregulated maturation states and aberrant antigen expression [37]. These characteristics make the dataset a natural testbed for analyzing distributional variation in Wasserstein space.

We used the publicly released BLAST110 dataset from Amsterdam UMC, which provides flow-cytometry measurements from 20 normal bone marrow (NBM) donors and 30 AML diagnosis (Dx) patients [38]. For this study, we focused on the P1 fluorescence panel, composed of eight immunophenotypic markers: CD7, CD56, CD34, CD117, CD33, HLA-DR, CD13, and CD45. These markers capture key aspects of hematopoietic differentiation and AML-related aberrancies, including progenitor signatures (CD34, CD117), myeloid lineage markers (CD13, CD33), and dysregulated antigen expression commonly observed in AML (CD7, CD56, HLA-DR).

Each sample contains on the order of hundreds of thousands of single-cell events. Because the dataset distributes pre-cleaned P1 files (including compensation and basic gating performed in the official workflow), our preprocessing incorporated only marker-level transformations. Fluorescence intensities were first transformed using the hyperbolic arcsine with cofactor $c = 5$,

$$x^{\text{trans}} = \text{arcsinh}(x/c),$$

followed by per-marker z -score normalization computed across the pooled cells of all samples. This procedure stabilizes variance across fluorescence channels and ensures comparability between samples.

To embed each sample into Wasserstein space, we represented its single-cell data as an empirical probability measure,

$$\mu_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \delta_{x_j}.$$

Direct computation of $\mathcal{W}_2$ on full-resolution flow-cytometry data is computationally prohibitive due to the large number of cells per sample. Consequently, for each of the 50 samples, we randomly selected $m = 1000$ cells without replacement and constructed

$$\mu_i^{(m)} = \frac{1}{m} \sum_{j=1}^m \delta_{x_j^{(i)}},$$

yielding $N = 50$ empirical distributions in $\mathcal{P}_2(\mathbb{R}^8)$. Subsampling of this scale has been shown to preserve major phenotypic structures in high-dimensional cytometry data while greatly reducing the cost of optimal-transport computations [39, 40].

We applied the Wasserstein K -median clustering algorithm with Partitioning Around Medoids (PAM) initialization and semi-free support IRLS median updates. For each $K \in \{2, 3, 4, 5, 6\}$, we alternated assignment and median-update steps until convergence. Cluster quality was assessed using the Wasserstein relative depth (ReD) statistic described in Section 2.4. The quantitative results are summarized in Table below.

The ReD statistic attains its maximum at $K = 2$, indicating that the most coherent cluster configuration corresponds precisely to the underlying biological division between NBM and AML Dx samples. By quantifying each distribution’s *relative centrality* within its assigned cluster, ReD is robust to inter-patient variability in blast fractions, marker intensity skewness, and shifts in differentiation

K	ReD (mean)
2	0.1604
3	0.1435
4	0.0812
5	0.0604
6	0.0586
Best K	2

state—factors that typically complicate distance-based clustering in flow-cytometry datasets. This outcome agrees with established AML biology, in which diagnosis samples display globally perturbed hematopoietic organization relative to healthy marrow. Consequently, the ReD-guided Wasserstein K -median framework provides a stable and interpretable characterization of population-level structure in this clinically relevant cohort.

6 Discussion and Future Work

This work combines semi-free support barycenters, Wasserstein Fréchet medians, and a K -center clustering strategy with a depth-based validation metric, ReD. The method performs reliably across synthetic mixtures, MNIST distributions, and AML flow cytometry data, and ReD consistently identifies meaningful cluster structure where distance-based validation metrics tend to overestimate K . On the theoretical side, we established a Pollard-type strong consistency result for Wasserstein K -means, providing one of the first guarantees linking empirical and population cluster centers in $\mathcal{P}_2(X)$.

Despite these encouraging numerical results and the fact that Pollard-style consistency extends to Wasserstein space under compact support assumptions, our analysis still does not guarantee that the barycenter computed at each step of the K -center iteration remains inside the “data-generated” family of measures. In general, the Wasserstein barycenter of a cluster may lie outside the observed sample and even outside the smallest natural class generated by it, and we currently lack structural conditions ensuring that all iterates stay within a geometrically meaningful subset determined by the underlying sampling model. Understanding when barycenters of empirical measures and the geodesics connecting them remain inside an admissible class of distributions thus remains an important open question for clarifying the domain of applicability of our algorithm.

Our framework implicitly assumes that clusters are approximately geodesically convex, a regime in which Wasserstein spatial depth induces a coherent center-outward ordering and ReD effectively distinguishes within- from between-cluster geometry. When clusters are nonconvex or multimodal, however, global depth may mask valleys in the Wasserstein landscape and lead to cluster merging. This suggests extending the approach to *local* Wasserstein depth, analogous to local depth in Euclidean spaces, where neighborhoods are defined by Wasserstein balls or transport-based connectivity graphs.

Another limitation is that ReD is a purely empirical validation metric. Unlike K -means clustering, no consistency result is known for selecting K via ReD or other depth-based criteria in Wasserstein space. Depth values depend on the global geometry of the empirical sample, and as the number of distributions grows, the between-cluster gap required for ReD to reliably distinguish clusters may shrink or become obscured by sampling noise. Establishing even weak or asymptotic guarantees for ReD—such as characterizing conditions under which the maximizer of its population analogue coincides with the true number of clusters—remains completely open. Because depth involves a geometric average of transport directions, its limiting behavior depends on geodesicity, curvature, and barycentric regularity of $\mathcal{P}_2(X)$, none of which are yet fully understood.

Several theoretical questions therefore remain open. In particular, while our consistency result assumes uniqueness of the population K -means solution, general sufficient conditions for uniqueness in Wasserstein space are unknown due to the nonlinearity and potential nonuniqueness of geodesics and barycenters. Developing such conditions, as well as theoretical guarantees for ReD-based model selection, are promising directions for future work. Computationally, future development will focus on multi-scale Sinkhorn schemes, low-rank transport approximations, and structure-aware support parameterizations to improve scalability in high-dimensional distributional applications.

References

- [1] Victor M. Panaretos and Yoav Zemel. “Statistical aspects of Wasserstein distances”. In: *Annual Review of Statistics and Its Application* 6 (2019), pp. 405–431. DOI: [10.1146/annurev-statistics-030718-104938](https://doi.org/10.1146/annurev-statistics-030718-104938).
- [2] François Bachoc et al. “Wasserstein spatial depth”. In: *arXiv preprint arXiv:2411.10646* (2024).
- [3] Dongkuan Xu and Yingjie Tian. “A comprehensive survey of clustering algorithms”. In: *Annals of data science* 2.2 (2015), pp. 165–193.
- [4] Yubo Zhuang, Xiaohui Chen, and Yun Yang. “Wasserstein K -means for clustering probability distributions”. In: *Advances in Neural Information Processing Systems* 35 (2022), pp. 11382–11395.
- [5] Zixiao Wang, Dong Qiao, and Jicong Fan. “Spectral clustering for discrete distributions”. In: *arXiv preprint arXiv:2401.13913* (2024).
- [6] Eustasio Del Barrio et al. “optimalFlow: optimal transport approach to flow cytometry gating and population matching”. In: *BMC bioinformatics* 21.1 (2020), p. 479.
- [7] Soheil Kolouri et al. “A continuous linear optimal transport approach for pattern analysis in image datasets”. In: *Pattern recognition* 51 (2016), pp. 453–462.
- [8] Jiacheng Zhu et al. “Functional optimal transport: regularized map estimation and domain adaptation for functional data”. In: *Journal of Machine Learning Research* 25.276 (2024), pp. 1–49.
- [9] Adam Jaeger and David Banks. “Cluster analysis: A modern statistical review”. In: *Wiley Interdisciplinary Reviews: Computational Statistics* 15.3 (2023), e1597.
- [10] Susanne Still and William Bialek. “How many clusters? An information-theoretic perspective”. In: *Neural computation* 16.12 (2004), pp. 2483–2506.
- [11] Kathleen M Gates et al. “Assessing the robustness of cluster solutions obtained from sparse count matrices.” In: *Psychological Methods* 24.6 (2019), p. 675.
- [12] Peter J Rousseeuw. “Silhouettes: a graphical aid to the interpretation and validation of cluster analysis”. In: *Journal of computational and applied mathematics* 20 (1987), pp. 53–65.
- [13] Yanchi Liu et al. “Understanding of internal clustering validation measures”. In: *2010 IEEE international conference on data mining*. Ieee. 2010, pp. 911–916.
- [14] C-H Chou, M-C Su, and Eugene Lai. “A new cluster validity measure and its application to image compression”. In: *Pattern Analysis and Applications* 7.2 (2004), pp. 205–220.
- [15] David L Davies and Donald W Bouldin. “A cluster separation measure”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* PAMI-1.2 (1979), pp. 224–227.
- [16] Mohammed J Zaki and Wagner Meira. *Data mining and analysis: fundamental concepts and algorithms*. Cambridge University Press, 2014.
- [17] Maria Halkidi and Michalis Vazirgiannis. “Clustering validity assessment: Finding the optimal partitioning of a data set”. In: *Proceedings 2001 IEEE international conference on data mining*. IEEE. 2001, pp. 187–194.
- [18] Robert Tibshirani, Guenther Walther, and Trevor Hastie. “Estimating the number of clusters in a data set via the gap statistic”. In: *Journal of the royal statistical society: series b (statistical methodology)* 63.2 (2001), pp. 411–423.
- [19] Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient flows: in metric spaces and in the space of probability measures*. Springer, 2005.
- [20] Davoud Moulavi et al. “Density-based clustering validation”. In: *Proceedings of the 2014 SIAM international conference on data mining*. SIAM. 2014, pp. 839–847.
- [21] Erich Schubert and Peter J Rousseeuw. “Faster k-medoids clustering: improving the PAM, CLARA, and CLARANS algorithms”. In: *International conference on similarity search and applications*. Springer. 2019, pp. 171–187.
- [22] Yijun Zuo and Robert Serfling. “General notions of statistical depth function”. In: *Annals of statistics* (2000), pp. 461–482.

- [23] Rebecka Jörnsten, Yehuda Vardi, and Cun-Hui Zhang. “A robust clustering method and visualization tool based on data depth”. In: *Statistical Data Analysis Based on the L1-norm and Related Methods*. Springer, 2002, pp. 353–366.
- [24] Ethan Anderes, Stefan Borgwardt, and Jacob Miller. “Discrete Wasserstein Barycenters: Existence, Uniqueness and Computation via Linear Programming”. In: *SIAM Journal on Scientific Computing* 38.1 (2016), A220–A242.
- [25] Marco Cuturi and Arnaud Doucet. “Fast computation of Wasserstein barycenters”. In: *International conference on machine learning*. PMLR, 2014, pp. 685–693.
- [26] Kisung You, Dennis Shung, and Mauro Giuffrè. “On the Wasserstein median of probability measures”. In: *Journal of Computational and Graphical Statistics* 34.1 (2025), pp. 253–266.
- [27] Thibaut Le Gouic and Jean-Michel Loubes. “Existence and consistency of Wasserstein barycenters”. In: *Annals of Statistics* 45.2 (2017), pp. 704–738.
- [28] Jérémie Bigot, Thierry Klein, et al. “Characterization of barycenters in the Wasserstein space by averaging optimal transport maps”. In: *Annals of Statistics* 45.2 (2017), pp. 957–990.
- [29] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi. “Optimization by Simulated Annealing”. In: *Science* 220.4598 (1983), pp. 671–680.
- [30] Vladimír Černý. “Thermodynamical approach to the traveling salesman problem: An efficient simulation algorithm”. In: *Journal of optimization theory and applications* 45.1 (1985), pp. 41–51.
- [31] Emile Aarts and Jan Korst. *Simulated Annealing and Boltzmann Machines: A Stochastic Approach to Combinatorial Optimization and Neural Computing*. Wiley, 1989.
- [32] Leonard Kaufman and Peter J. Rousseeuw. *Finding Groups in Data: An Introduction to Cluster Analysis*. Wiley, 1987.
- [33] Regina Y Liu. “On a notion of data depth based on random simplices”. In: *The Annals of Statistics* (1990), pp. 405–414.
- [34] David Pollard. “Strong consistency of k-means clustering”. In: *The annals of statistics* (1981), pp. 135–140.
- [35] Peter Gänssler and Winfried Stute. “Empirical processes: a survey of results for independent and identically distributed random variables”. In: *The Annals of Probability* (1979), pp. 193–243.
- [36] Yvan Saeys, Sofie Van Gassen, and Bart N Lambrecht. “Computational flow cytometry: helping to make sense of high-dimensional immunology data”. In: *Nature Reviews Immunology* 16.7 (2016), pp. 449–462.
- [37] Junqing Wu et al. “A single-cell survey of cellular hierarchy in acute myeloid leukemia”. In: *Journal of hematology & oncology* 13.1 (2020), p. 128.
- [38] Tim R Mocking et al. “Computational assessment of measurable residual disease in acute myeloid leukemia using mixture models”. In: *Communications Medicine* 4.1 (2024), p. 271.
- [39] Nikolay Samusik et al. “Automated mapping of phenotype space with single-cell data”. In: *Nature methods* 13.6 (2016), pp. 493–496.
- [40] Jacob H Levine et al. “Data-driven phenotypic dissection of AML reveals progenitor-like cells that correlate with prognosis”. In: *Cell* 162.1 (2015), pp. 184–197.